# Open Items Register

Technical Design Document — Cubli

Armonia — Space Challenges 2026

Scan of August 8, 2026

## How to use this document

This register lists every unresolved item in the TDD: the TBD markers, the sections that are empty, the subsections that exist only as comments, and the figures that are referenced but not drawn. It is a working checklist, not part of the submitted document.

Three status labels are used throughout:

- **READY** — decidable now. Nothing upstream is missing; the item is waiting on a decision or a datasheet lookup, not on data.
- **WAITING** — blocked on an upstream item, named in the row.
- **BLOCKER** — blocks a large number of downstream items. Close these first.
- **CLOSED** — resolved since the original scan. Retained in the register so the closure is auditable rather than silently deleted.

Section **??** gives the section-file map for the current draft. Section 1 states the dependency chain, which is what makes the ordering non-arbitrary: roughly a quarter of the register closes automatically once the packaging study closes.

## 1 Section-file map (current draft)

The former `03_analysis.tex` (RW-vs-CMG trade, requirements traceability) and `06_testing.tex` (excluded from the build) are retired; their content is redistributed into the files above rather than existing as standalone sections. Section labels most items are still addressed by (`sec:massbudget`, `sec:jumpup_target`, `sec:rw_sizing`, `sec:control2d`, `sec:test_feasibility` → `sec:sizing_verification`, `sec:test_electrical` → `sec:elec_checks`, `sec:test_cad` → `sec:cad_verification`, `sec:test_sim` → `sec:sim_environment`) are unchanged or carried forward under a new name; the label, not the file, is the stable identifier to search on.

## 2 Dependency chain

Most items are not independent. The governing chain is

$$V_{\min} \to L \to M \to h_w \to I_{w,\text{target}} \to \text{wheel design, feasibility gate, test thresholds}$$

so the packaging study (item **A4**, Table **??** in `10_cad_detail.tex`) is the true head of the register. The parametric calculator `analysis/SIZING.py` implements Stages 1–6 of the closure procedure (Section **??**, now in `03_preliminary_sizing.tex`) and reports a converged fixed point at $M = 1.7716\,\text{kg}$, reproduced by its own Stage 6 roll-up to within 0.1 g. The figures below are **transcribed into the TDD** (A1, A2 and A8 are closed). They remain provisional because roughly 34 % of the mass total is still structural allowance rather than measured mass.

| # | Section | File |
|---|---|---|
| 1 | Introduction (incl. RW-vs-CMG trade) | `01_introduction.tex` |
| 2 | Project Organisation (incl. validation stage-gate table) | `02_project_organization*.tex` |
| 3 | Preliminary Sizing | `03_preliminary_sizing.tex` |
| 4 | Methodology (incl. verification/checks) | `04_methodology.tex` |
| 5 | CAD Design (incl. CAD verification) | `05_cad_design.tex` |
| 6 | Electronics | `06_electronics.tex` |
| 7 | Control Algorithms & Approach | `07_control_algorithms.tex` |
| 8 | Conclusions & Next Steps (incl. validation results, future work) | `08_conclusions.tex` |
| A | Bill of Materials | `09_bom.tex` |
| B | CAD & Construction Detail | `10_cad_detail.tex` |
| C | Electrical System Detail (schematic sheets, now uncommented, plus per-pin tables) | `11_electrical_detail.tex` |
| D | Control Algorithm Detail (state machine, pseudocode, gains, timing, faults) | `12_control_algorithm_detail.tex` |

**What remains open on the sizing chain.** The residual risk is not a conflict but a margin asymmetry: the wheel sits 11.7 % above target while one station is worth 5.9 %, so downward ballast trim is nearly exhausted (−1 station) while upward trim is ample (+24). A structure that comes in *lighter* than the 610 g allowance is therefore the awkward case, and $\omega_{\max}$ is the relief variable. This is unchanged by the restructure — the numbers moved files, not values.

**Table 1:** Converged values from `analysis/SIZING.py`, as transcribed into the TDD. The wheel diameter is pinned by the packaging envelope (`D_w_FIXED_mm` = 120), not chosen by the mass-minimising sweep.

| Symbol | Quantity | Converged value | Status |
|---|---|---|---|
| $M$ | Cube mass (Stage 1 assumption) | 1.7716 kg | Fixed point |
| $M_{\text{est}}$ | Cube mass (Stage 6 roll-up) | 1.7716 kg | $\Delta \approx 0$; 34% allowance |
| $L$ | Cube edge length | 150 mm | Fixed |
| $\eta$ | Energy margin | 1.05 | Chosen |
| $\tau_b$ | Brake torque per wheel | 5 N m | Assumed |
| $\tau_g = MgL/2$ | Gravitational tip-over floor | 1.303 N m | Derived |
| $\beta$ | Brake-amplification factor | 1.353 | Derived |
| $\omega_{\max}$ | Max wheel speed | 6000 rpm | Agreed with TDD |
| $h_w$ | Required angular momentum | $0.2588\,\mathrm{kg\,m^2\,s^{-1}}$ | Derived |
| $I_{w,\text{target}}$ | Target wheel inertia (corner) | $4.119 \times 10^{-4}\,\mathrm{kg\,m^2}$ | Derived |
| $I_{w,\text{target}}$ | Target wheel inertia (edge) | $3.737 \times 10^{-4}\,\mathrm{kg\,m^2}$ | Not critical |
| $D_w$ | Wheel outer diameter | 120 mm | Envelope constraint |
| $N$ | Ballast stations per wheel | 15 of $N_{\max} = 39$ | Derived |
| $m_w$ | Wheel mass (each) | 172.5 g | Derived |
| $I_w$ | Wheel inertia achieved | $4.599 \times 10^{-4}\,\mathrm{kg\,m^2}$ | 111.7 % of target |

# 3 TBD items, by owning table

## 3.1 A1 — Jump-up sizing budget CLOSED

`tab:jumpup_budget`, `03_preliminary_sizing.tex` (`sec:jumpup_target`). All 6 values populated from the converged run of Table 1; provisional in the sense that $M$ carries a 34 % structural allowance.

## 3.2 A2 — Required inertia by contact case CLOSED

`tab:inertia_cases`, `03_preliminary_sizing.tex` (`sec:massbudget`). Both contact cases populated: edge $I_{w,\text{target}} = 3.737 \times 10^{-4}\,\text{kg}\,\text{m}^2$; corner $4.119 \times 10^{-4}$. The corner case demands 9.3 % more inertia.

## 3.3 A3 — Rail current budget READY

`tab:power_budget`, `06_electronics.tex` (`sec:power_arch`). **14 TBD**: typical and worst case for each of five loads plus the logic-rail total and the driver spin-up figure. This table was previously wrapped in a LaTeX `comment` environment and did not render at all, despite being referenced from Appendix C; it is now a live table with the same TBD content. Typical values are datasheet lookups and can be filled today; worst case is the *coincident* case (telemetry burst, servo stall and full sensor load together), not the sum of typicals.

## 3.4 A4 — Driving CAD parameters BLOCKER

`tab:cad_params`, `10_cad_detail.tex`. **6 TBD, unchanged**: $d_m$, $h_m$, $c_w$, $c_o$, $V_{\min}$, $r_c$. These need the packaging study, not the calculator. $V_{\min}$ is step 2 of the closure procedure and therefore the head of the whole dependency chain: it fixes $L$, which fixes $M$, which fixes $I_{w,\text{target}}$. $c_w$ and $c_o$ are additionally consumed by the interference checks of `sec:cad_verification` in `05_cad_design.tex`.

## 3.5 A5 — CAD model tree and part list READY

`tab:cad_parts`, `10_cad_detail.tex`. **34 TBD**: fabrication route and status for every part across the inner structure, wheel assembly, outer frame and 1-DoF rig, unchanged from the previous scan. All are decidable today and do not depend on the sizing.

## 3.6 A6 — Print and manufacturing settings READY

`tab:print_settings`, `10_cad_detail.tex`. **18 TBD**: material, infill/walls and orientation for the six structural printed parts (W1, W2, M1, M2, F1, F2). The calculator assumes PET-CF at $1290\,\text{kg}\,\text{m}^{-3}$, which should be the material entry for W1 at minimum.

## 3.7 A7 — Mass properties verification WAITING

`tab:mass_verification`, `10_cad_detail.tex`. **12 TBD**: CAD value, measured value and discrepancy for $M$, $m_w$, $I_w$ and CoM offset. The CAD column can be filled once the model exists; the measured column requires assembled hardware.

## 3.8 A8 — Pre-fabrication feasibility gate CLOSED

`tab:jumpup_gate`, `03_preliminary_sizing.tex` (`sec:sizing_verification`, formerly `sec:test_feasibility`). All 8 values populated and the gate passes on every row: speed margin 0.896; inertia route disagreement 11.7 %; ballast trim $-1/+24$ stations; retention at 1.47 % of proof load.

### 3.9 A9 — 1-DoF plant identification WAITING

`tab:paramid_methods`, `04_methodology.tex` (`sec:paramid`). **4 TBD**: torque constant $K_t$, body inertia $J$, friction, loop latency. Methods and instruments are specified; only the measured values and tolerances are pending. Requires the rig.

### 3.10 A10 — Control performance metrics WAITING

`tab:perf_metrics`, `08_conclusions.tex` (`sec:test_perf`). **5 TBD**: targets for recovery angle, settling time, disturbance-rejection bandwidth, slew tracking error, momentum drift. Control effort is already defined by reference to the driver limits.

### 3.11 A11 — Connector schedule READY

`tab:connectors`, `11_electrical_detail.tex`. **11 TBD**: housing and keying for J1–J11, unchanged. No two connectors carrying different voltages may share a housing type, so a mis-mate is impossible by construction; J1 is fixed by the BOM (XT60).

### 3.12 A12 — Wire schedule READY

`tab:wire_schedule`, `11_electrical_detail.tex`. **14 TBD**: gauge and length for H1–H7, unchanged. Gauge follows from A3; length follows from the CAD envelope.

### 3.13 A13 — As-built electrical measurements WAITING

`tab:electrical_measured`, `11_electrical_detail.tex`. **7 TBD**: $K_m$, $I_w$ from spin-down, viscous and Coulomb friction, command-to-encoder latency, achievable wheel speed on the bus, worst-case logic rail load, runtime per charge. Requires hardware.

### 3.14 A14 — Per-sheet pin/net logic tables READY / WAITING(mixed)

`tab:pins_power_entry` through `tab:pins_servo`, `11_electrical_detail.tex`, Sheets 1–7. **New since the restructure**: uncommenting the schematic-sheet placeholders and adding a per-pin electrical/logic table to each (net name, direction, level, function) surfaced **9 TBD** physical-pin-number entries (E-stop/fuse pin positions, the LM2596 pin-out, the CAN termination component refs, the Teensy CAN-FD peripheral pins). The net identity, direction and level in every row are fixed by the architecture of `sec:electrical` and are *not* TBD; only the physical pin number is, and it is **READY** in the sense that it needs a board layout decision (`sec:layout`), not upstream data — but that decision itself has not been made, so it is listed **WAITING** on it.

### 3.15 A15 — Appendix D: gains, timing budget, guard conditions WAITING

`12_control_algorithm_detail.tex`. **New since the restructure**: Appendix D was 0 bytes at the previous scan (see Section **??** note) and is now written in full — the mode- transition guard conditions (`tab:state_transitions`, 5 TBD), the fixed gains and tuning parameters (`tab:gains`, 10 TBD: $Q$, $R$, $K_d$ for both the edge and corner models, $\gamma$, $\alpha$, $\theta_{\text{th}}$, the desaturation threshold), the per-cycle timing budget (`tab:timing_budget`, 12 TBD across five pipeline stages), and two fault-response timeouts (`tab:fault_responses`, 2 TBD). All of these close only once the plant identification (A9) and the Simulink/Simscape simulation work of `sec:sim_environment` report measured parameters and re-derived gains — they are a consequence of A9 closing, not an independent blocker.

| Location | Pending |
|---|---|
| `08_conclusions.tex`, `sec:test_edge` | Edge-balance numeric targets beyond the G6 gate criterion. |
| `08_conclusions.tex`, `sec:test_slew` | Slew tracking targets. |
| `07_control_algorithms.tex`, `sec:sim_environment` | Robustness-sweep perturbation magnitude — set once the identification tolerances of A9 are known. |

# 4 TBD in prose

# 5 Empty sections

**None.** Both previously-empty content blocks are closed: Appendix D (`12_control_algorithm_detail.tex`) now carries the mode state machine, estimator and controller pseudocode, gains table, timing budget and fault-response table (residual TBDs tracked as A15); and the former Section 3 (RW-vs-CMG trade, requirements traceability) is folded into the Introduction and the opening of Preliminary Sizing respectively, so there is no longer a stub section for it.

Three headings remain deliberately empty, by standing decision rather than by omission — they are commented-out placeholders reserved for content not yet written, per the placeholder convention used throughout the document (numbered heading + label + one-line note, no filler prose):

| Section | File | State |
|---|---|---|
| NMPC & warm-start control | `08_conclusions.tex` | Heading only (commented) |
| Autonomous stand-up | `08_conclusions.tex` | Heading only (commented) |
| Single-wheel variant | `08_conclusions.tex` | Heading + comment |

# 6 Comment-only stubs

Down from twenty at the previous scan to one: the seven electrical schematic-sheet placeholders (`11_electrical_detail.tex`) are now live `\phfig` figures with captions and per-pin tables (A14), and the former Section 3 stubs no longer exist as a section.

| Location | Content |
|---|---|
| `05_cad_design.tex`, `app:cad_3d` in `10_cad_detail.tex` | 3D cube drawings: exploded view; section view through two orthogonal wheel modules. Drawing set not yet released — waits on A4. |

# 7 Missing figures

- `04_methodology.tex`, `sec:2dmodel` — free-body diagram of the 2D pendulum, labelling $\theta_b$, $\theta_w$, $l$, $l_b$ and the pivot. Unchanged from the previous scan; this is still the only figure suggested but not present in the dynamics section.
- `05_cad_design.tex`, `app:cad_3d` — cube exploded view and the section view named in the comment-only stub above.

The seven electrical schematic sheets are no longer in this list: they render as placeholder-art figures (Section 2, A14) rather than as comments, so they are tracked as artwork-pending rather than as missing figures.

## 8 Defects found alongside the TBDs

Historical record, retained for audit. All rows below predate the restructure; renumbered section/table references are current.

| # | Location | Defect |
|---|---|---|
| 1 | Power architecture vs. BOM | **CLOSED** Battery capacity contradiction resolved against the procured pack: Tattu R-Line V5.0, 22.2 V, 1550 mA h, 150C, XT60. |
| 2 | `tab:rw_params` | **CLOSED** Missing `\caption` added; now appears in the List of Tables. |
| 3 | G6 hard-kill date vs. `tab:gates` | **CLOSED** Both now state 17 August. |
| 4 | `02_project_organization_tasks` | Task B4 says "Raspberry Pi software architecture"; the rest of the document specifies Teensy 4.1 throughout. |
| 5 | BOM vs. CAD part list | Brake servo quantity: BOM says 1–3, the CAD part list says 3 brackets. |
| 6 | Author list vs. role/task tables | **CLOSED** Name spelling unified to "Niccolo" throughout. |
| 7 | `02_project_organization_tasks` | Task IDs C11–C14 appear under the "3D Design" block, after D1–D7. |
| 8 | All section files | British/American spelling mixed: realise/realize, characterise/characterize, behaviour/behavior. |
| 9 | **CLOSED** Hard-coded `\newpage` | Removed with the section it sat in during the restructure. |
| 10 | All tables | **CLOSED** Every float now uses `[htbp]`; no `[h]` float remains in `sections/`. |
| 11 | `sections/` | `code.cpp` and `code.exe` are untracked in the repo; a compiled binary should not be committed. |

## 9 Suggested order of work

This mirrors, and should be kept in sync with, the Immediate Next Steps subsection of `08_conclusions.tex` (`sec:next_steps`).

1. **Close the packaging study** (A4) — still the head of the register. $V_{\text{min}}$ and the clearances release $L$, then $M$. A1, A2 and A8 are populated but *provisional*: they rest on a 610 g structural allowance, so closing A4 moves them again as the next turn of the fixed-point iteration, not a re-opening of settled items.
2. **Build and run the Simulink/Simscape environment** (`sec:sim_environment`) against both prototypes' CAD, and re-derive the gains of A15 once A9 (plant ID) reports measured parameters.
3. **Do the READY items**, which need no upstream data: A3 (rail budget), A5 (fabrication routes), A6 (print settings), A11 (connectors), A12 (wire gauges).
4. **Decide the board layout** (`sec:layout`) to close the physical pin numbers of A14.
5. **Fix the remaining defects** (Section 2 above), which cost credibility out of proportion to the effort of fixing them.
6. **The remaining WAITING items** (A7, A9, A10, A13, A15 and the three prose TBDs) close on hardware and cannot be accelerated on paper.